

Solution Requirements

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Analyst/staff log in with username & password before accessing the fraud-review dashboard	High
2	Authorization	Role-based access: Analysts can review/flag; Admins can retrain/redeploy the model	Medium
3	External interfaces	Integrates with IBM Watson Machine Learning REST API for hosted model scoring	High
4	Transactions / data processing	Accepts a transaction's Time, Amount and V1–V28 features and returns a fraud probability in real time	High
5	Reporting	Dashboard shows flagged transactions, model confidence, and daily fraud-rate summary	Medium
6	Business rules, etc.	Transactions above a configurable fraud-probability threshold are auto-flagged for review	High
7	Compliance to laws or regulations	Handles cardholder data per PCI-DSS-style data-minimization principles (no raw PAN/CVV stored)	High
8	Other	CSV batch-upload mode for compliance officers to screen many transactions at once	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric
1	Performance & Speed	Model must return a prediction almost instantly for a single transaction	Response < 2 seconds
2	Scalability	System should handle spikes in transaction volume without degrading	Supports 1,000+ req/min via cloud scaling
3	Security & Data Privacy	Sensitive data encrypted in transit; PCA features already anonymize raw card data	HTTPS/TLS + no raw PAN storage
4	Reliability & Availability	Scoring service should stay up during business hours	99% uptime target
5	Usability & Accessibility	Simple form-based UI so a non-technical analyst can use it	Task completion without training
6	Other	Model and code should be easy to retrain as new fraud patterns emerge	Maintainable modular pipeline